

Introduction

This project explores how AI-driven non-playable characters (NPCs) can mimic social interactions in crowded environments. We utilize Unreal Engine's Environmental Query System (EQS), Blueprints and Behavior trees to achieve this. By focusing on crowd density and environmental factors, we aim to replicate realistic behaviors where NPCs adjust their speed, elect alternate routes, and respond to social cues dynamically. Through this, our research seeks to enhance the believability of virtual simulations, creating a powerful tool for applications in game development, urban planning, and beyond, where human interaction within crowd dynamics is crucial.

Abstract Summary

Goal:

Investigate how AI non-playable characters (NPCs) can be used to simulate human-like social interactions in crowded environments. Specifically, we are focusing on how AI-driven NPCs dynamically adjust their movement and interactions based on crowd density, and various environmental factors.

This is Accomplished:

Utilizing the Environmental Query System (EQS), NPCs can perceive environmental factors within a certain area. They generate random personality traits to determine how said factors will affect their actions. Using the EQS, the NPC will generate a set of points, grade each point based on the

scoring system, and then select a point to move to.

NPC movement is tracked using a heatmap system, and all data is saved to a custom storage where it can be exported as a CSV file for analysis.

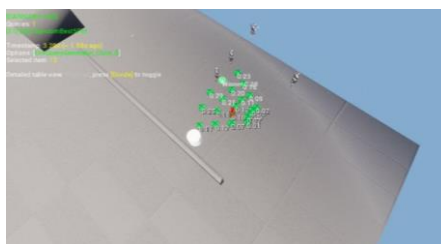


Figure A: Visualized EQS for one NPC persona

Research Progress



Location outputting and logging: This component involves tracking the location of each NPC as they navigate the virtual environment. Each NPC is assigned an ID, allowing us to monitor and differentiate their movements. Using C++ scripts, we record the NPC's coordinates at regular intervals, saving the data to a CSV file for further analysis.



Variable Speed: To better simulate realistic human beings, we implemented a system to ensure that each NPC is spawned with a randomly generated maximum walking speed. This is done through the blueprints system and a random number generator with a user defined range.



Despawning: To manage crowd sizes and resolve NPCs reaching their goals, a collision box is used to remove a NPC once it has reached its goal.. The blueprint system triggers despawning when an NPC overlaps with the collision box surrounding the goal.

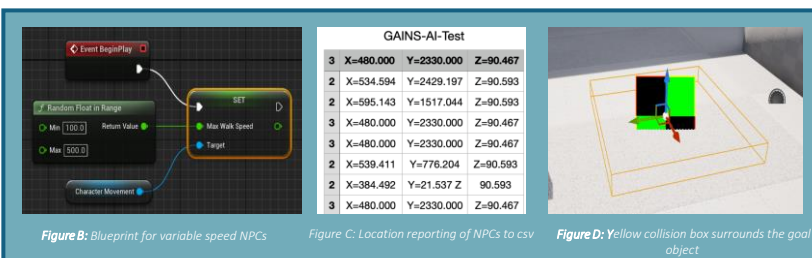


Figure B: Blueprint for variable speed NPCs

Figure C: Location reporting of NPCs to csv

Figure D: Yellow collision box surrounds the goal object

Crowd Generation:

Crowd generation within the project involves creating spawn volumes in the environment to facilitate the random placement of NPCs. This method ensures diverse crowd distributions that replicate natural human movement. Additionally, individual NPCs can be spawned using custom spawners, providing precise control over specific scenarios. These spawn systems integrate seamlessly with the EQS and Persona system, enabling NPCs to react dynamically to environmental stimuli based on their unique profiles. This setup ensures that NPCs exhibit varied and realistic responses to elements like sound and crowd sizes, contributing to a more authentic simulation.

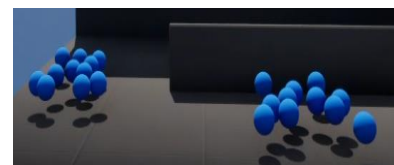


Figure E: Spawn volumes placed in the environment

Sound Emission in Unreal Engine:

- This component simulates sound propagation using Unreal Engine 5's *Line Trace by Channel* function. Each line trace models a sound path, dynamically adjusting its direction upon contact with surfaces to represent reflections. The system also simulates sound absorption by reducing the trace's intensity by a variable percentage with each surface interaction, mimicking how surfaces dampen sound. This enhances the overall simulation by providing realistic sound behavior in the environment and allows NPCs to more accurately react and make decisions based off sound.

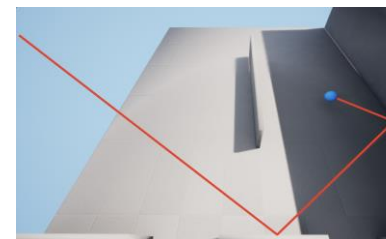


Figure F: Demonstration of Sound Reflection

Why is this research important?

Simulating human-like behavior through predictive modeling with NPCs can bridge future practical applications such as improving AI in complex and dynamic systems. By modeling crowded environments, and the way NPCs handle unpredictable conditions, which helps enhance AI systems as they become an integral part of everyday life in a modern world. Research surrounding human characteristics and decision-making abilities. Using predictive simulations allows for preemptive actions to take place, when applied beyond the modeling system.

Future Plans

We aim to integrate the sound emission and trace system into the NPC decision-making process. When a sound trace reaches an NPC, it will trigger context-specific responses, such as moving toward the sound, retreating from it, or performing another action based on the NPC's perception of the sound. These behaviors will be tailored to each NPC's unique personality, with varying priorities and reactions to environmental stimuli, enhancing the realism and complexity of their interactions. In addition, we plan on having the NPCs continuously change their walking speeds based on their surroundings. An example could be a NPC slowing down in a crowd of people or speeding up to pass a large group. This would allow us to more accurately reflect crowd behavior. Another thing to consider would be if the NPCs were able to set secondary goals to further the chance of intermingling and inter-NPC interactions.

References

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